

A Tower-Recursive Source for Every Gold Diagonal

a9lim

August 9, 2026

Abstract

Let m be a power of two, let $K_m = \mathbb{F}_{2^m}$ be the canonical finite nim-field, and write $e_i = 2^i$ for its Conway bit basis. For the Gold form

$$Q_a(x) = \text{Tr}_{K_m/\mathbb{F}_2}(x^{1+2^a}),$$

the diagonal constants $Q_a(e_i)$ are the datum missing from the polar form. Previously a uniform game-built source was known only for $a = 1$, while the trace-dual elements for even a visibly drifted under tower doubling. We give a recursion for the trace-dual diagonal of every scaled Gold component, prove that the unscaled dual always descends to the half-field, and conclude that it is an Artin–Schreier value $w^2 + w$. Thus every Gold exponent has a single tower-constructed source w satisfying

$$\text{Tr}((w^2 + w)e_i) = Q_a(e_i) \quad (0 \leq i < m).$$

The result closes the diagonal-sourcing problem, including the even- a case. By itself it does not turn those constants into a normal-play rule; the companion weighted-source Witt–FIFO theorem in `writeups/goldarf.tex` supplies that separate strategic step.

1 Statement and boundary

For a scale $c \in K_m$, define the scaled Gold component

$$Q_{a,c}^{(m)}(x) = \text{Tr}_m(cx \text{Fr}^a(x)), \quad \text{Fr}(x) = x^2,$$

where $\text{Tr}_m = \text{Tr}_{K_m/\mathbb{F}_2}$. Its canonical-basis diagonal extends uniquely to an $\mathbb{F}_2$ -linear functional

$$\ell_{a,c}^{(m)}\left(\sum_i x_i e_i\right) = \sum_i x_i Q_{a,c}^{(m)}(e_i).$$

The absolute trace pairing on a finite separable field is nondegenerate, so there is a unique $\Lambda_{a,c}^{(m)} \in K_m$ such that

$$\text{Tr}_m(\Lambda_{a,c}^{(m)} e_i) = \text{Tr}_m(c e_i \text{Fr}^a(e_i)) \quad \text{for every } i < m. \quad (1)$$

Write $\Lambda_a^{(m)} = \Lambda_{a,1}^{(m)}$.

Theorem 1.1 (all-exponent Gold-diagonal source). *For every power of two $m \geq 2$ and every $a \geq 0$:*

1. $\Lambda_{a,c}^{(m)}$ is given by the quadratic-tower recursion in Theorem 2.2, using only nim addition and multiplication, Frobenius, relative trace, and the canonical tower generator;

2. the unscaled dual descends:

$$\Lambda_a^{(m)} \in K_{m/2} \subset K_m;$$

3. there is $w_a^{(m)} \in K_m$ with

$$\Lambda_a^{(m)} = \wp(w_a^{(m)}) = (w_a^{(m)})^2 + w_a^{(m)};$$

and therefore

$$Q_a(e_i) = \text{Tr}_m\left(\wp(w_a^{(m)})e_i\right) \quad (0 \leq i < m).$$

The two possible sources are $w_a^{(m)}$ and $w_a^{(m)} + 1$.

The theorem is about sourcing the diagonal constants. It does not itself infer a P -set theorem or a linking strategy. Those are supplied by the separate weighted-source Witt–FIFO theorem; this proof/semantics boundary is essential even though the combined project-level problem is now resolved.

2 The quadratic nim-tower block

Put $m = 2M$. The canonical nim tower has

$$K_m = K_M(u) = K_M \oplus uK_M, \quad u = e_M,$$

and the nontrivial K_M -automorphism $\sigma = \text{Fr}^M$ satisfies

$$\sigma(u) = u + 1.$$

The canonical basis splits as

$$e_0, \dots, e_{M-1}, \quad ue_0, \dots, ue_{M-1};$$

indeed $ue_j = e_{M+j}$. Let

$$T = T_{m/M} = 1 + \sigma : K_m \longrightarrow K_M$$

be the relative trace.

Lemma 2.1 (trace-pairing blocks). *For $A, B, e \in K_M$,*

$$T((A + uB)e) = Be, \tag{2}$$

$$T((A + uB)ue) = (A + B)e. \tag{3}$$

Proof. Because σ fixes A, B, e and sends u to $u + 1$,

$$(A + uB)e + \sigma((A + uB)e) = (A + uB)e + (A + (u + 1)B)e = Be.$$

For the upper block, apply the same calculation to $(A + uB)u$:

$$\begin{aligned} & (A + uB)u + (A + (u + 1)B)(u + 1) \\ &= Au + u^2B + Au + A + (u + 1)^2B = A + B, \end{aligned}$$

since characteristic two cancels the doubled Au , u^2B , and uB terms. Multiplication by e gives (3). $\square$

Now write $c = c_0 + uc_1$ and set

$$d_0 = T(c), \quad d_1 = T(cu \operatorname{Fr}^a(u)). \quad (4)$$

For a lower basis vector $e \in K_M$,

$$Q_{a,c}^{(m)}(e) = \operatorname{Tr}_M(T(c)e \operatorname{Fr}^a(e)) = Q_{a,d_0}^{(M)}(e).$$

For its upper mate ue ,

$$Q_{a,c}^{(m)}(ue) = \operatorname{Tr}_M(T(cu \operatorname{Fr}^a(u))e \operatorname{Fr}^a(e)) = Q_{a,d_1}^{(M)}(e).$$

Thus d_0 and d_1 are exactly the smaller-field scales of the two diagonal blocks.

Theorem 2.2 (scaled dual recursion). *Let*

$$L_0 = \Lambda_{a,d_0}^{(M)}, \quad L_1 = \Lambda_{a,d_1}^{(M)},$$

with d_0, d_1 as in (4). Then

$$\Lambda_{a,c}^{(2M)} = (L_0 + L_1) + uL_0. \quad (5)$$

The base case over $K_1 = \mathbb{F}_2$ is $\Lambda_{a,c}^{(1)} = c$.

Proof. Write the candidate in (5) as $A + uB$, where $B = L_0$ and $A = L_0 + L_1$. By Lemma 2.1, its relative trace pairing with a lower basis vector e is L_0e , while its pairing with ue is

$$(A + B)e = (L_0 + L_1 + L_0)e = L_1e.$$

After applying Tr_M , these are exactly the two defining dualities for the lower and upper diagonal blocks. Nondegeneracy of the absolute trace pairing makes the candidate unique, proving (5). $\square$

This is the point at which the even- a drift ceases to be an obstruction. The scale d_1 can change at every tower level; the recursion transports that change instead of requiring one fixed dual element throughout the tower.

3 Descent and the Artin–Schreier source

Lemma 3.1 (unscaled half-field descent). *For $m = 2M$,*

$$\Lambda_a^{(m)} \in K_M.$$

Proof. In the unscaled case $c = 1$, the lower scale is

$$d_0 = T(1) = 1 + 1 = 0.$$

Linearity (or the recursion from the base case) gives $L_0 = \Lambda_{a,0}^{(M)} = 0$. Theorem 2.2 therefore reduces to

$$\Lambda_a^{(2M)} = L_1 \in K_M.$$

$\square$

Lemma 3.2 (Artin–Schreier exact sequence). *For a finite field $K = \mathbb{F}_{2^m}$,*

$$\text{im}(x \mapsto x^2 + x) = \ker(\text{Tr}_{K/\mathbb{F}_2}).$$

Each element of this common hyperplane has exactly two preimages, differing by 1.

Proof. The Artin–Schreier map $\wp(x) = x^2 + x$ is $\mathbb{F}_2$ -linear and has kernel $\mathbb{F}_2 = \{0, 1\}$, so its image has $|K|/2$ elements. Frobenius invariance of trace gives

$$\text{Tr}(\wp(x)) = \text{Tr}(x^2) + \text{Tr}(x) = \text{Tr}(x)^2 + \text{Tr}(x) = 0,$$

so the image lies in the trace-zero hyperplane. The finite-field trace is a nonzero $\mathbb{F}_2$ -linear functional, hence its kernel also has $|K|/2$ elements. The inclusion is equality. Every nonempty fibre is a coset of the two-element kernel. $\square$

Proof of Theorem 1.1. Theorem 2.2 proves the constructive first assertion and Lemma 3.1 proves the second. If $\lambda = \Lambda_a^{(2M)} \in K_M$, trace transitivity gives

$$\text{Tr}_{2M}(\lambda) = \text{Tr}_M(\text{Tr}_{K_{2M}/K_M}(\lambda)) = \text{Tr}_M(2\lambda) = 0.$$

Lemma 3.2 supplies w with $w^2 + w = \lambda$. Substituting this identity into the defining trace duality (1) yields the stated formula for every basis coin. The fibre statement in Lemma 3.2 gives the two sources w and $w + 1$. $\square$

4 Implementation and formal boundary

The Rust core implements the proof literally:

- `gold_component_diagonal_dual(m,a,c)` evaluates Theorem 2.2;
- `gold_diagonal_dual(m,a)` specializes to $c = 1$; and
- `gold_diagonal_artin_schreier_source(m,a)` uses the existing exact finite-nim-field solver to choose one of the two sources.

The implementation is checked directly against all defining trace-pairing equations for every scale over K_m at $m \leq 8$. For the unscaled theorem it checks half-field descent, the Artin–Schreier equation, and every diagonal coordinate at representative exponents through the full $m = 128$ backend. The previous even- a values

$$\Lambda_2^{(4)} = 0, \quad \Lambda_2^{(8)} = 6, \quad \Lambda_2^{(16)} = 102, \quad \Lambda_2^{(32)} = 24582$$

and $\Lambda_4^{(16)} = 31$, $\Lambda_4^{(32)} = 8030$ are pinned as regression values; their drift is now explained and handled by the scale recursion.

The Lean module `formal/Ogdoad/GoldDiagonal.lean` kernel-checks the quadratic trace-block formulas, the dual-block reconstruction, the vanishing of the absolute trace of a quadratic-extension base element, additivity of the Artin–Schreier map, the one-step tower lift of a source, and the full finite-field equivalence $\text{Tr}(\lambda) = 0 \iff \lambda = w^2 + w$. The last result is proved from rank-nullity, trace surjectivity, and the two-element kernel, rather than introduced as an axiom. The module does not encode the finite nim tower or replace the Rust implementation with a Lean finite-field model.

5 Consequence for the resolved Gold-game program

The former even- a constructive sourcing question is closed: for every exponent, the diagonal is supplied by one deterministically selected tower constant through nim multiplication, Frobenius, trace, and finally one square plus XOR at the source identity $\Lambda = \wp(w)$. No coordinate-by-coordinate evaluation of Q_a occurs in the recursion. Unlike the $a = 1$ Fermat-coin formula, the final w is selected by exact finite-field linear algebra; the theorem does not claim a comparably simple closed expression or that this selection itself satisfies the strategic naturality axioms.

The strategic layer is resolved separately by weighted-source Witt-FIFO. Its source-pair weights consume these diagonal bits locally, the matching theorem forces the exact charge in every dimension, and the phase-aware claim lantern gives an ordinary normal-play P-set $\{Q_a = 0\}$. The present theorem supplies the all- a algebraic input to that construction; it does not collapse the distinction between sourcing and play semantics.